M50 auditory sound setup

Parts rationale wiring and verification
16 October 2026

We plan to build a Mac sound system with an Earthworks M50, Audient iD14 MKII and Edifier MR3. The first checkout will use the mouse setup, then the audio equipment will move to sheep. The M50 is the selected microphone; iD14 is the recommended interface upgrade. MR3 remains a budget speaker candidate until acoustic tests pass.

Our proposed sound target is approximately 1.4 to 32 kHz, with consecutive 10 ms sine pips and 2 ms ramps at each end. We will play and record at 96 kHz. Neural recording and clock integration remain later work. [S1]

Parts to purchase

Item and buying link	Qty	Unit	Line total	Role
Earthworks M50	1	$1,299.00	$1,299.00	Measurement mic
Audient iD14 MKII	1	$349.00	$349.00	Playback and capture
Edifier MR3 powered pair	1	$149.99	$149.99	Sound delivery
REED R8090 standard	1	$219.00	$219.00	1 kHz level reference
Hosa HMIC-010 XLR 10 ft	1	$17.99	$17.99	Mic to interface
Hosa HSS-005 TRS 5 ft	2	$14.95	$29.90	Speaker and reference
Gator RI-MICTP-FBM boom stand	1	$34.99	$34.99	Microphone placement
Listed parts total			$2,099.87	

Prices were checked on 16 October 2026. All seven listings indicated stock; none is reserved. The MR3 price is for a pair, and the TRS quantity is two cables.

Budget: $2,099.87 in parts + $50 for small support or USB needs + $200 tax and shipping reserve = $2,349.87. Set aside about $2,400. The allowances are estimates, not a delivered quotation or an upper bound.

Included: M50 microphone clip and ADP1 calibrator adapter; iD14 USB C to USB C cable; R8090 9 V battery. MR3 has its own amplifier. We use the existing Mac. [S8, S10, S12]

Also confirm a stable speaker shelf or support, a ruler or tape measure, position marks, a mains outlet and several GB of Mac storage. A small clip thread adapter or different USB cable may be needed. The $50 allowance assumes simple mounting or reuse; specialized mounts would add cost.

Why we chose the microphone and speaker



M50 on the left and MR3 pair on the right. Manufacturer product photographs; source records are retained in web/assets/parts/sources.json.

Earthworks M50

We chose the M50 because the measurement requirement extends above normal audio frequencies. Its specified 3 Hz to 50 kHz band covers our 32 kHz endpoint, and each microphone has its own response chart and electronic calibration data. This gives us a documented way to correct the microphone response when measuring the highest tones. Its XLR output connects through the Mac audio interface. [S2, S12]

The higher microphone cost buys a measurement reference that we can own and reuse for mouse and sheep work. Register the delivered microphone, request its serial matched electronic calibration file, and inspect that file through at least 32 kHz. Keep the original data and its correction convention. Factory sensitivity at 1 kHz and frequency response correction are separate pieces of calibration. [S3, S12]

Edifier MR3

We chose the MR3 as the inexpensive speaker to evaluate. It includes amplification, accepts a balanced TRS line signal, and has a nominal 52 Hz to 40 kHz range. The manufacturer response illustration also supports output near 32 kHz, rather than relying only on a printed upper frequency. A single documented cabinet/channel keeps the initial sound field simple. [S4, S5]

The remaining question is usable output at our level and geometry. The response illustration does not specify all test conditions, and published independent measurements end near 20 kHz. Neither source establishes consecutive 10 ms pips at 32 kHz. The manual rates power with dynamic range compression on; its effect on our sequence needs measurement. Keep MR3 only if the actual sequence meets the lab's level and timing criteria. [S5, S6, S14]

Why the cheaper mic was excluded

The Dayton EMM-6 has individual calibration data but is specified only to 20 kHz. It could support an audible subset, but cannot establish our full 32 kHz calibration requirement. The M50 choice preserves the broader sound target. [S13]

Why we chose the interface and level reference

Audient iD14 MKII

The iD14 plays the waveform and records two signals at once: the M50 acoustic signal and a duplicate electrical waveform. Two rear microphone or line inputs support that arrangement. Its preamp at minimum gain and ADC are specified through 40 kHz; its DAC is specified through 40 kHz. It supplies 48 V phantom power at 10 mA per channel, matching the documented M50 power requirement. [S2, S7]

This is why we recommend iD14 over the earlier iD4. The iD4 can perform the original acoustic checkout, but has only one full band mic or line input. The iD14 upgrade and second TRS cable add $93.96 to the previous basket and give us a stronger timing measurement path. The revised list includes one interface. Actual working gain, channel matching and native Mac playback/capture still need checks. [S7, S15]

At 96 kHz, each 10 ms pip contains 960 samples. This is a functional Rice adaptation from the paper's 192 kHz DAC. The fundamental at 32 kHz is below the 48 kHz Nyquist limit, but the recording cannot capture its 64 kHz second harmonic. We can measure the required fundamentals without claiming complete ultrasonic distortion coverage. [S1, S7]

REED R8090

The calibrator supplies a known 94 or 114 dB signal at 1 kHz. With the M50 and a fixed interface gain, it establishes the conversion from recorded sample amplitude to sound pressure. The M50 correction file describes frequency response; the R8090 checks the recording level scale. Its stated accuracy is +/-0.5 dB, and its half inch port nominally matches the included ADP1 adapter. Verify the seal. [S10, S12]

Start with the 94 dB setting and inspect headroom. The 114 dB setting can clip at a gain used for stimulus recording. The $219 standard model does not include the separately priced traceable calibration certificate. A 1 kHz reference does not independently establish accuracy at 32 kHz or certify the entire measurement chain.

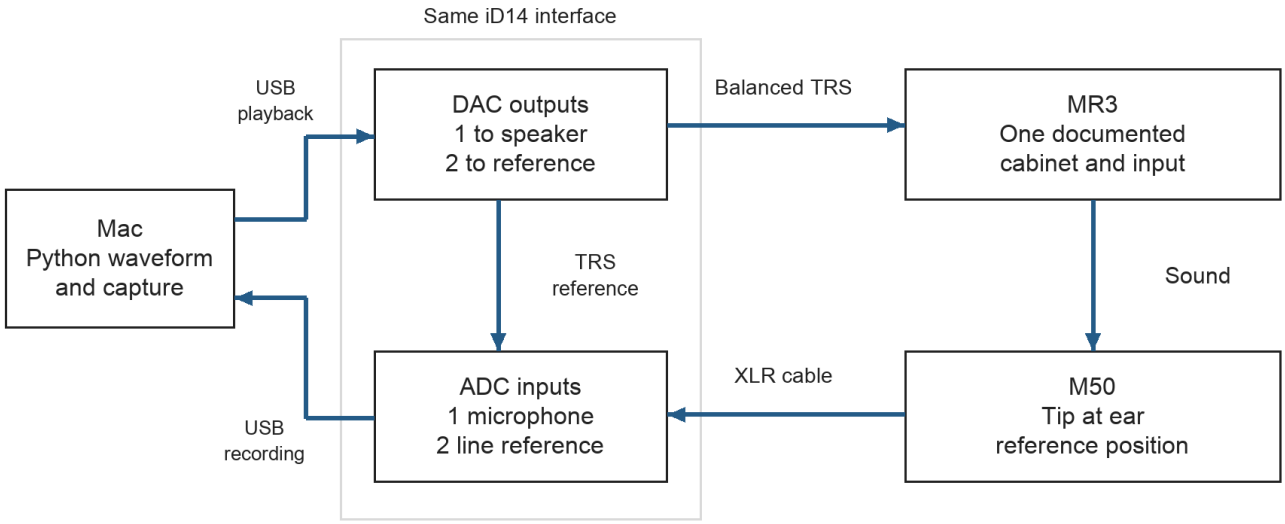
Cables and positioning

The XLR cable carries the microphone signal and phantom power. One balanced TRS cable carries sound to the MR3; the second carries the electrical reference back to the interface. The boom stand holds the microphone tip at a repeatable reference position. It is a microphone stand, not the speaker support. Confirm the supplied clip thread fits before relying on the setup.

The Mac USB connection must provide at least 0.9 A at 5 V. Prefer a direct suitable USB port with the included cable. Save the exact Mac and audio settings after confirming native 96 kHz duplex operation. [S7, S8]

Software to configure: Audient iD software for interface routing, and Python with NumPy and sounddevice for waveform generation and playback/capture. Save recordings and metadata to writable Mac storage. Native operation on the chosen Mac must be checked after installation. [S8, S9, S11]

How we will wire and position the equipment



Signal connections for simultaneous sound and electrical reference recording

Connection	Part	Setting or check
Mac to iD14	Included USB C cable	Native 96 kHz playback and capture
Output 1 to MR3	Balanced TRS cable 1	Verify the input drives the chosen cabinet
Output 2 to rear input 2	Balanced TRS cable 2	Line input with phantom off and low gain
M50 to rear input 1	XLR cable	Microphone input with phantom on
R8090 over M50 tip	Included ADP1 adapter	Separate level check then remove calibrator

Leave the front instrument input empty because it takes over channel 1. Mute both recorded inputs from every speaker/output mix to prevent feedback. Route the intended playback channels explicitly. DAW Thru bypasses iD Mixer level controls, so reduce source amplitude before selecting it and verify which controls affect the actual output. [S8, S9]

Place the selected MR3 cabinet on a stable support and align its tweeter with the ear/reference position. Place the M50 tip at that position for the acoustic checkout, pointing as required by its free field calibration. Measure and record distance, height and orientation; photograph the arrangement and mark positions. The mouse distance is still undecided.

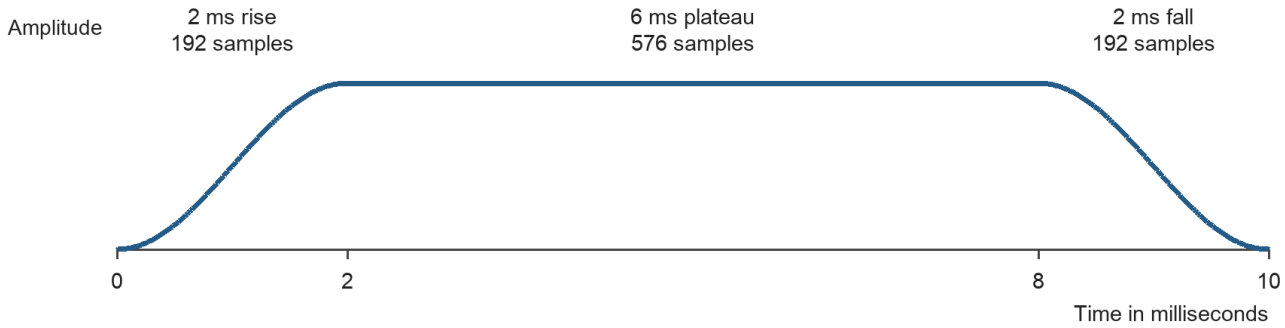
Use wired TRS, Monitor mode and fixed documented EQ/room settings. Record firmware and standby settings; follow the manufacturer's standby guidance. The second DAC output is a duplicate reference. Characterize relative DAC/ADC channel gain and delay before treating it as a timing reference for output 1. [S4, S7 to S9]

How we will program the sound

Python on the Mac creates sample arrays. The interface converts them to voltage at a fixed sample rate; the speaker converts voltage to sound. We prepare each continuous block before playback so Python does not schedule individual tones using 10 ms sleeps.

Select the iD14 explicitly in Python. Route macOS alerts and other applications away from it, and prevent computer sleep during a run. Save the actual device and channel mapping.

Parameter	Planned value	Meaning
Sample rate	96,000 samples per second	Waveform sampling resolution
Tone duration	10 ms or 960 samples	100 consecutive pips per second
End ramps	2 ms or 192 samples each	Included inside the 10 ms pip
Between ramps	6 ms or 576 samples	Central plateau
Frequency band	Approximately 1.4 to 32 kHz	Final frequency list remains a lab choice



Illustrative raised cosine envelope for one 10 ms pip at 96 kHz

The paper specifies zero starting phase and 2 ms end ramps but does not specify the ramp shape. A raised cosine is our proposed implementation choice. The exact original 37 frequency array is absent from the methods; do not present a reconstructed list as that published array. [S1]

Software sequence

1. Build a balanced list of the chosen frequencies, shuffle using a saved seed, and generate each sine pip using the appropriate calibrated amplitude.
2. Join the pips without silence between them. For timing tests, retain leading and trailing silence so acoustic onset and final decay are captured.
3. Duplicate the waveform to playback outputs 1 and 2. Record microphone input 1 and electrical reference input 2 at the same native sample rate.
4. Use sounddevice playrec for short initial checks and an explicit continuous duplex stream for controlled longer runs. Inspect underflow/overflow status. [S11]

Save the frequency schedule, digital onset sample numbers, random seed, amplitude table, exact waveform or its reproducible source, raw two channel recording and all device/geometry settings. Keep raw data alongside any corrected result. A 30 minute stereo float32 capture at 96 kHz is about 1.38 GB; stream blocks to disk rather than requiring the whole session in memory.

Calibration and the tests that qualify the setup

Establish the measurement scale

1. Register the M50 and request its serial matched ECF. Inspect coverage through 32 kHz, units and whether values describe microphone response or corrective gain. Confirm the correct interpretation before applying it. [S2, S3]
2. Set a microphone gain, fit the ADP1 and R8090, and record the 94 dB reference. Check the seal and avoid clipping. Remove the calibrator for free field measurement. Keep the gain fixed; recalibrate after any gain change. [S10, S12]
3. At the empty ear/reference position, record background sound and test tones across the selected frequencies. Apply the microphone correction and fixed gain level scale to estimate sound pressure. Check the interface response at the operating settings as well.
4. Build a per frequency playback amplitude table that reaches the selected level within measured speaker headroom. Define whether SPL refers to the central plateau or RMS across the whole ramped pip; use the same convention for calibration and validation.

Measure the actual sequence

Check	Measurement	Pass definition
Frequency and level	All retained tones including 27 and 32 kHz	Lab level and tolerance at fixed geometry
Short pip behavior	Actual 10 ms pips and large frequency jumps	Duration spacing and residual energy tolerances
Timing reference	Microphone versus characterized electrical reference	Delay and variability within lab criteria
Stability	Repeated complete blocks and before/after level checks	No unexplained missing events clipping or drift

Paper blocks contain 8,000 to 15,000 pips, lasting 80 to 150 seconds. At least 12 blocks imply about 16 to 30 minutes plus 1 second block gaps. This is a useful paper style duration test, not an automatically approved mouse protocol. [S1]

A fixed playback delay longer than 10 ms can still preserve a 100 pip per second sequence. Frequency dependent delay, compression and ringing need acoustic recordings. Sample arithmetic and long steady tones alone do not establish this behavior.

Keep the MR3 only after it meets the agreed acoustic criteria. A quick mouse checkout is the first stage. When moving the equipment to sheep, repeat level/frequency calibration and sequence checks at the new geometry. Neural marker/clock alignment must be designed separately before neural timing claims.

Decisions and source references

The M50 is selected. iD14 and the extra reference cable are the recommended verification revision. The remaining decisions belong to the experimental team, and the current list has not passed a physical acoustic test.

Decision	Current status	Owner
Mouse distance and orientation	TBD before acoustic checkout	Howard and lab
Mouse level and SPL convention	TBD under the lab protocol	Lab lead with Howard
Frequency list and ramp shape	TBD; raised cosine proposed	Howard and lab
Numerical level and timing tolerances	TBD before judging measurements	Lab lead with Howard
Sheep geometry and neural alignment	Later stage; recalibration required	Howard Jiaao and lab

SONIC reports awake, alert sheep and approximately 80 dB SPL at sheep ear distance. Planned Rice sheep sedation is a team adaptation. The paper level is not a mouse exposure prescription, and neither the hardware list nor an acoustic checkout establishes neural performance. [S1]

Sources

S1 SONIC v1 methods. Saved PDF printed p 11 Sections 6.1 to 6.3; p 13 Section 6.6.

S2 Earthworks M50 specifications. Bandwidth power load and individual calibration.

S3 Earthworks serial calibration request. Request the file for the delivered microphone.

S4 Edifier MR3 specifications and delay FAQ. Range inputs processing firmware and standby.

S5 Edifier MR3 response illustration. Supports 32 kHz output under unspecified conditions.

S6 Erin original MR3 measurements. Independent measurements end near 20 kHz.

S7 Audient iD14 MKII specifications. Inputs outputs power and frequency response.

S8 Audient connections and phantom power. Independent phantom switches and front DI behavior.

S9 Audient output routing. DAW Thru and monitoring configuration.

S10 REED R8090 specifications. Level accuracy frequency port and battery.

S11 Python sounddevice playback and recording. Initial playrec checks and continuous streams.

S12 Earthworks M50 user guide. Included clip adapter and calibration information.

S13 Dayton EMM-6 specifications. 18 Hz to 20 kHz and individual calibration file.

S14 Edifier MR3 user manual. Monitor mode and power rating with DRC on.

S15 Audient iD4 MKII specifications. Earlier one full band input comparison.

Buying links are embedded in the parts table on page 1. Manufacturer product photographs are reused from the project's recorded image sources. Wiring and envelope figures are explanatory schematics, not experimental measurements.